If we can prove that $|x-z_0|, |x_0-y| \le L$, then we should have

$$(x-x_0)\cdot(y-z_0) \ge 0$$

and we would use this to get a contradiction, by means of a suitable choice for ξ. Indeed, the direction of $x-x_0$ is almost that of ξ (up to an error of δ) and that of $y-z_0$ is almost that of y_0-z_0 (up to an error of the order of $\varepsilon/|z_0-y_0|$). If we choose ξ such that $\xi\cdot(y_0-z_0)<0$, this means that for small δ and ε we would get a contradiction.

We need to guarantee $|x-z_0|, |x_0-y| \le L$, in order to prove the claim. Let us compute $|x-z_0|^2$: we have

$$|x-z_0|^2 = |x_0-z_0|^2 + |x-x_0|^2 + 2(x-x_0)\cdot(x_0-z_0).$$

In this sum we have

$$|x_0-z_0|^2 \le L^2; \quad |x-x_0|^2 \le r^2; \quad 2(x-x_0)\cdot(x_0-z_0) = |x-x_0|(\xi\cdot(x_0-z_0)+O(\delta)).$$

Suppose now that we are in one of the following situations: either choose ξ such that $\xi\cdot(x_0-z_0)<0$ or $|x_0-z_0|^2<L^2$. In both cases we get $|x-z_0|\le L$ for r and δ small enough. In the first case, since $|x-x_0| \ge \frac{r}{2}$, we have a negative term of the order of r and a positive one of the order of r^2; in the second we add to $|x_0-z_0|^2<L^2$ some terms of the order of r or r^2.

Analogously, for what concerns $|x_0-y|$ we have

$$|x_0-y|^2 = |x_0-x|^2 + |x-y|^2 + 2(x_0-x)\cdot(x-y).$$

The three terms satisfy

$$|x_0-x|^2 \le r^2; \quad |x-y|^2 \le L^2;$$
$$2(x_0-x)\cdot(x-y) = |x-x_0|(-\xi\cdot(x_0-y_0)+O(\delta+\varepsilon+r)).$$

In this case, either we have $|x_0-y_0|<L$ (which would guarantee $|x_0-y|<L$ for ε,δ small enough), or we need to impose $\xi\cdot(y_0-x_0)<0$.

Note that imposing $\xi\cdot(y_0-x_0)<0$ and $\xi\cdot(x_0-z_0)<0$ automatically gives $\xi\cdot(y_0-z_0)<0$, which is the desired condition so as to have a contradiction. Set $\mathbf{v}=y_0-x_0$ and $\mathbf{w}=x_0-z_0$. If it is possible to find ξ with $\xi\cdot\mathbf{v}<0$ and $\xi\cdot\mathbf{w}<0$ we are done. When is it the case that two vectors $\mathbf{v}$ and $\mathbf{w}$ do not admit the existence of a vector ξ with both scalar products that are negative? The only case is when they go in opposite directions. But this would mean that x_0, y_0 and z_0 are colinear, with z_0 between x_0 and y_0 (since we supposed $|x_0-z_0|\le|x_0-y_0|$). If we want z_0 and y_0 to be distinct points, we should have $|x_0-z_0|<L$. Hence in this case we do not need to check $\xi\cdot(x_0-z_0)<0$. We only need a vector ξ satisfying $\xi\cdot(y_0-x_0)<0$ and $\xi\cdot(y_0-z_0)<0$, but the directions of y_0-x_0 and y_0-z_0 are the same, so that this can be guaranteed by many choices of ξ, since we only need the scalar product with one only direction to be negative. Take for instance $\xi=-(y_0-x_0)/|y_0-x_0|$. □ □

Theorem 3.24. *The secondary variational problem*

$$\min\{K_2(\gamma) : \gamma\in O_\infty(\mu,\nu)\}$$

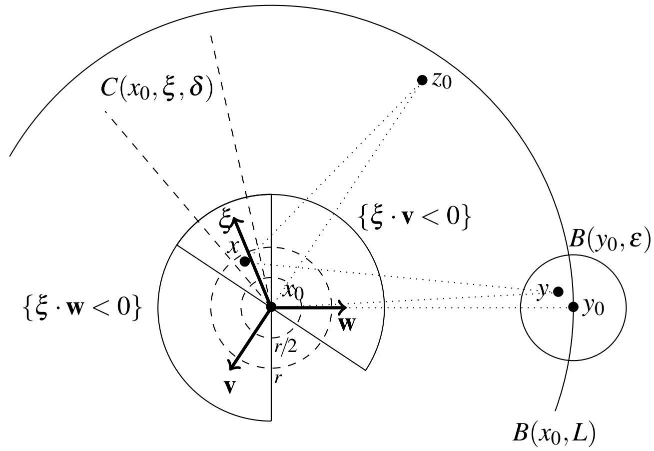

Figure 3.4: The points in the proof of Lemma 3.23.

admits a unique solution $\overline{\gamma}$*, it is induced by a transport map* T*, and such a map is an optimal transport for the problem*

$$\min\{||\mathrm{T} - \mathrm{id}||_{L^\infty(\mu)}\ ,\ \mathrm{T}_\#\mu = \nu\}.$$

Proof. We have already seen that $\overline{\gamma}$ is concentrated on a set $\widetilde{\Gamma}$ satisfying some useful properties. Lemma 3.23 shows that $\widetilde{\Gamma}$ is contained in a graph, since for any x_0 there is no more than one possible point y_0 such that $(x_0, y_0) \in \widetilde{\Gamma}$. Let us consider such a point y_0 as the image of x_0 and call it $\mathrm{T}(x_0)$. Then $\overline{\gamma} = \gamma_{\mathrm{T}}$. The optimality of T in the "Monge" version of this L^∞ problem comes from the usual comparison with the Kantorovich version on plans γ. The uniqueness comes from standard arguments (since it is true as well that convex combinations of minimizers should be minimizers, and this allows to perform the proof in Remark 1.19). □ □

3.3 Discussion

3.3.1 Different norms and more general convex costs

The attentive reader will have observed that the proof in Section 3.1 about the case of the distance cost function was specific to the case of the distance induced by the Euclidean norm. Starting from the original problem by Monge, and relying on the proof strategy by Sudakov [290] (which had a gap later solved by Ambrosio in [8, 11], but was meant to treat the case of an arbitrary norm), the case of different norms has been extensively studied in the last years. Note that the case of uniformly convex norms (i.e. those such that the Hessian of the square of the norm is bounded from below by a matrix which is positively definite) is the one which is most similar to the Euclidean case, and can be treated in a similar way.

One of the first extension was the case studied by Ambrosio, Kirchheim and Pratelli about crystalline norms ([19]), i.e. norms such that their unit balls are convex polyhedra. One can guess that the fact that we have a finite number of faces makes the task easier, but it already required a huge approximation work. Indeed, if one uses duality, the gradient $\nabla u(x)$ of a Kantorovich potential is not enough to determine the direction of the displacement $y-x$ for a pair $(x,y) \in \mathrm{spt}(\gamma)$, because a crystalline norm has the same gradient on all the points of a same face (and not only on points on the same ray, as is the case for the Euclidean norm, or for other strictly convex norms). To handle this problem, [19] develops a strategy with a double approximation, where $||z||$ is replaced by something like $||z|| + \varepsilon|z| + \varepsilon^2|z|^2$, where $||z||$ is the norm we consider, $|z|$ is the Euclidean norm which is added so as to select a direction, and its square is added so as to get strict convexity and monotonicity on each ray (exactly as we saw in Section 3.1). In this way the authors prove the existence of an optimal map for crystalline norms in any dimensions, and for arbitrary norms in $\mathbb{R}^2$.

Later, the way was open to the generalization to other norms. The main tool is always a secondary variational problem, since for norm costs it is in general false that optimal plans all come from a map, and one needs to select a special one. The difficulty with secondary variational problems, is that they correspond to a new cost c which is not finitely valued. This prevents from using duality on the secondary problem. Indeed, Proposition 1.42 proves that we have equality between the primal and the dual problem for all costs which are l.s.c., but does not guarantee the existence of an optimizer in the dual; existence is usually guaranteed either via Ascoli-Arzelà arguments, or through Theorem 1.37, which requires finiteness of the cost. This is why some methods avoiding duality, and concentrating on c-cyclical monotonicity have been developed. The first time they appeared is in [123], for a different problem (an L^∞ case): it corresponds to the idea that we presented in Section 3.2 (looking for density points of this concentration set). Later, Champion and De Pascale managed to use the same tools to prove the existence of an optimal map first for arbitrary strictly convex norms (in [120], the same result being obtained differently at almost the same time by Caravenna in [104]), and then for general norms in [121]. This last result was more difficult to obtain and required some extra approximation tools inspired from [273], and in particular the selection of a special optimal transport plan via approximation through atomic marginals, as we did in Section 3.1.5.

But the history of optimal transport did not look only at norms. Many studies have been done for different distance cost functions (distances or squared distances on manifolds, geodesic distances on $\mathbb{R}^d$ if obstacle are present...). In this book we prefer to stick to the Euclidean case, and in this section we only consider costs of the form $c(x,y) = h(x-y)$ for h convex. Even this case is far from being completely understood. In a paper with Carlier and De Pascale, see [113], a general (straightforward) strategy of decomposition according to the "faces" of the cost function h is presented.

The decomposition is based on the following steps:

- Consider an optimal plan γ and look at the optimality conditions as in Section 1.3. For all $(x_0, y_0) \in \mathrm{spt}(\gamma)$, if x_0 is a differentiability point for the potential φ (we write $x_0 \in \mathrm{Diff}(\varphi)$), one gets $\nabla\varphi(x_0) \in \partial h(x_0 - y_0)$, which is equivalent to

$$x_0 - y_0 \in \partial h^*(\nabla\varphi(x_0)). \tag{3.6}$$

Let us define

$$\mathscr{F}_h := \{\partial h^*(p) \ :\ p \in \mathbb{R}^d\},$$

which is the set of all possible values of the subdifferential multi-map of h^*. These values are those convex sets where the function h is affine, and they will be called *faces* of h. The fact that φ is differentiable μ-a.e. is enforced by supposing $\mu \ll \mathscr{L}^d$ and h to be Lipschitz, so that φ is also Lipschitz.

Thanks to (3.6), for every fixed x, all the points y such that (x,y) belongs to the support of an optimal transport plan are such that the difference $x-y$ belongs to a same face of c. Classically, when these faces are singletons (i.e. when c^* is differentiable, which is the same as c being strictly convex), this is the way to obtain a transport map, since only one y is admitted for every x.

Equation (3.6) also enables one to classify the points x as follows. For every $K \in \mathscr{F}_h$ we define the set

$$X_K := \{x \in \mathrm{Diff}(\varphi) \ :\ \partial h^*(\nabla\varphi(x)) = K\}.$$

Hence γ may be decomposed into several subplans γ_K according to the criterion $x \in X_K$. If K varies among all possible faces this decomposition covers γ-almost all pairs (x,y). Moreover, if (x,y) belongs to $\mathrm{spt}(\gamma)$ and x to $\mathrm{Diff}(\varphi)$, then $x \in X_K$ implies $x-y \in K$.

If the set $\mathscr{F}_h$ is finite or countable, we define

$$\gamma_K := \gamma_{|X_K \times \mathbb{R}^d}.$$

In this case, the marginal measures μ_K and ν_K of γ_K (i.e. its images under the maps π_x and π_y, respectively) are sub-measures of μ and ν, respectively. In particular μ_K inherits the absolute continuity from μ. This is often useful for proving existence of transport maps.

If $\mathscr{F}_h$ is uncountable, in some cases one can still rely on a countable decomposition by considering the set $\mathscr{F}_h^{multi} := \{K \in \mathscr{F}_h \ :\ K \text{ is not a singleton }\}$. If $\mathscr{F}_h^{multi}$ is countable, then one can separate those x such that $\partial h^*(\nabla(\varphi(x))$ is a singleton (where a transport already exists) and look at a decomposition for $K \in \mathscr{F}_h^{multi}$ only.

- This decomposition reduces the transport problem to a superposition of transport problems of the type

$$\min\left\{\int h(x-y)\,\mathrm{d}\gamma(x,y) \ :\ \gamma \in \Pi(\mu_K,\nu_K),\ \mathrm{spt}(\gamma) \subset \{x-y \in K\}\right\}.$$

The advantage is that the cost c restricted to K is easier to study. If K is a face of h, then h is affine on K and in this case the transport cost does not depend any more on the transport plan.

- The problem is reduced to find a transport map from μ_K to ν_K satisfying the constraint $x - \mathrm{T}(x) \in K$, knowing a priori that a transport plan satisfying the same constraint exists.

In some cases (for example if K is a convex compact set containing 0 in its interior) this problem may be reduced to an L^∞ transport problem. In fact if one denotes by $||\cdot||_K$ the (gauge-like) "norm" such that $K = \{x : ||x||_K \leq 1\}$, one has

$$\min\left\{\max\{||x-y||_K,\,(x,y)\in \mathrm{spt}(\gamma)\},\gamma\in\Pi(\mu,\nu)\right\}\leq 1 \tag{3.7}$$

and the question is whether the same inequality would be true if one restricted the admissible set to transport maps only (passing from Kantorovich to Monge, say). The answer would be yes if a solution of (3.7) were induced by a transport map T. This is what we presented in Section 3.2 in the case $K = \overline{B(0,1)}$, and which was first proven in [123] via a different strategy (instead of selecting a minimizer via a secondary variational problem, selecting the limit of the minimizers for the L^p norms or $p \to \infty$). Note also that this issue is easy in 1D, where the monotone transport solves all the L^p problems, and hence the L^∞ as well (and this does not need the measure to be absolutely continuous, but just atomless).

Let us note that the assumption that the number of faces is countable is quite restrictive, and is essentially used to guarantee the absolute continuity of μ_K, with no need of a disintegration argument (which could lead to difficulties at least as hard as those faced by Sudakov). However, an interesting example that could be approached by finite decomposition is that of crystalline norms. In this case the faces of the cost h are polyhedral cones but, if the support of the two measures are bounded, we can suppose that they are compact convex polyhedra. This means, thanks to the considerations above, that it is possible to perform a finite decomposition and to reduce the problem to some L^∞ minimizations for norms whose unit balls are polyhedra (the faces of the cone). In particular the L^1 problem for crystalline norms is solved if we can solve L^∞ optimal transport problem for other crystalline norms.

It becomes then interesting to solve the L^∞ problem as in Section 3.2, replacing the unit ball constraint $|x-y|\leq 1$ with a more general constraint $x-y\in C$, the set C being a generic convex set, for instance a polyhedron. This is studied in [198] by minimizing the quadratic cost

$$c(x,y)=\begin{cases}|x-y|^2 & \text{if } x-y\in C,\\ +\infty & \text{otherwise,}\end{cases}$$

which is also of the form $c(x,y) = h(x-y)$ for h strictly convex (but not real valued). The existence of an optimal map (better: the fact that any optimal γ for this problem is induced by a map) is proven when μ is absolutely continuous and C is either strictly convex or has a countable number of faces. This can be achieved by adapting the arguments of Section 3.2 and proving that, if (x_0,y_0) and (x_0,y_1) belong to $\mathrm{spt}(\gamma)$ (and hence $y_0-x_0, y_1-x_0\in C$), then the middle point $\frac{1}{2}(y_0+y_1)-x_0$ cannot lie in the interior of C. If C is strictly convex this proves $y_0=y_1$. If not, this proves that the points y_0-x_0 and y_1-x_0 are on a same face of C and one can perform a dimensional reduction argument, and proceed by induction.

In particular, this argument applies to the case of polyhedra and of arbitrary convex sets in $\mathbb{R}^2$ (since no more than a countable quantity of segments may be contained in ∂C), and provides an alternative proof for the result of [19].

Many generalizations of this last result have been performed: Bertrand and Puel ([54]) studied the "relativistic" cost $h(z) := 1 - \sqrt{1-|z|^2}$, which is naturally endowed with the constraint $|z| \le 1$ and is a strictly convex function (but not finite-valued since the constraint means $h(z) = +\infty$ outside the unit ball). They proved existence of an optimal map by adapting the arguments of [123, 198] to the case of a strictly convex function h with a strictly convex constraint C (from there, adapting to the case of a countable number of faces seems to be also possible).

Finally, Chen, Jiang and Yang combined the arguments of [198] with those of [121], thus getting existence for the cost function

$$c(x,y) = \begin{cases} |x-y| & \text{if } x-y \in C, \\ +\infty & \text{otherwise,} \end{cases}$$

under the usual assumptions on C (strictly convex or countable number of faces). These results can be found in [124, 125] and are the first which combine lack of strict convexity and infinite values.

All these subsequent improvements paved the way to an obvious conjecture, i.e. the fact that a mixture of duality, variational approximation and density points techniques could finally prove the existence of an optimal transport map in a very general case:

Problem (existence for every convex cost): suppose $\mu \ll \mathscr{L}^d$ and take a cost $c(x,y)$ of the form $h(x-y)$ with $h : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ convex; prove that there exists an optimal map.

This problem was open till the final redaction phases of this book, when Bardelloni and Bianchini paper [27] appeared, containing a proof of a disintegration result implying this very same statement, at least in the case where h is finite-valued but not strictly convex.

3.3.2 Concave costs (L^p, with $0 < p < 1$)

Another class of transport costs which is very reasonable for applications, rather than convex functions of the Euclidean distance, is that of concave costs[3], more precisely $c(x,y) = \ell(|x-y|)$ where $\ell : \mathbb{R}_+ \to \mathbb{R}_+$ is a strictly concave and increasing function[4]. From the economic and modeling point of view, this is the most natural choice: moving a mass has a cost which is proportionally less if the distance increases, as everybody can note from travel fares of most transport means (railways, highway tolls, flights...). All these costs are sub-additive. In many practical cases, moving two masses, each on a distance d, is more expensive than moving one at distance $2d$ and keeping at rest the other. The typical example of costs with these property is given by the power cost $|x-y|^\alpha$, $\alpha < 1$.

Note that all these costs satisfy the triangle inequality and are thus distances on $\mathbb{R}^d$. Moreover, under strict concavity assumptions, these costs satisfy a strict triangle inequality. This last fact implies that the common mass between μ and ν must stay at rest, a fact first pointed out in [177].

[3] We mean here costs which are concave functions of the distance $|x-y|$, not of the displacement $x-y$, as instead we considered in Remark 2.12.

[4] We will not explicitly state it every time, but this also implies that ℓ is strictly increasing.

Theorem 3.25. *Let γ be an optimal transport plan for the cost $c(x,y) = \ell(|x-y|)$ with $\ell : \mathbb{R}_+ \to \mathbb{R}_+$ strictly concave, increasing, and such that $\ell(0) = 0$. Let $\gamma = \gamma_D + \gamma_O$, where γ_D is the restriction of γ to the diagonal $D = \{(x,x) : x \in \Omega\}$ and γ_O is the part outside the diagonal, i.e. the restriction to $D^c = (\Omega \times \Omega) \setminus D$. Then this decomposition is such that $(\pi_x)_\#\gamma_O$ and $(\pi_y)_\#\gamma_O$ are mutually singular measures.*

Proof. It is clear that γ_O is concentrated on $\mathrm{spt}(\gamma) \setminus D$ and hence $(\pi_x)_\#\gamma_O$ is concentrated on $\pi_x(\mathrm{spt}(\gamma) \setminus D)$ and $(\pi_y)_\#\gamma_O$ on $\pi_y(\mathrm{spt}(\gamma) \setminus D)$. We claim that these two sets are disjoint. Indeed suppose that a common point z belongs to both. Then, by definition, there exists y such that $(z,y) \in \mathrm{spt}(\gamma) \setminus D$ and x such that $(x,z) \in \mathrm{spt}(\gamma) \setminus D$. This means that we can apply c-cyclical monotonicity to the points (x,z) and (z,y) and get

$$\ell(|x-z|) + \ell(|z-y|) \le \ell(|x-y|) + \ell(|z-z|) = \ell(|x-y|) < \ell(|x-z|) + \ell(|z-y|),$$

where the last strict inequality gives a contradiction. □ □

This gives a first constraint on how to build optimal plans γ: look at μ and ν, take the common part $\mu \wedge \nu$, leave it on place, subtract it from the rest, and then build an optimal transport between the two remainders, which will have no mass in common. Note that when the cost c is linear in the Euclidean distance, then the common mass *may* stay at rest but is not forced to do so (think at the book-shifting example); on the contrary, when the cost is a strictly convex function of the Euclidean distance, in general the common mass *does not* stay at rest (in the previous example only the translation is optimal for $c(x,y) = |x-y|^p$, $p > 1$). Note that the fact that the common mass stays at rest implies that in general there is no optimal map T, since whenever there is a set A with $\mu(A) > (\mu \wedge \nu)(A) = \nu(A) > 0$ then almost all the points of A must have two images: themselves, and another point outside A.

A typical case is represented in Figure 3.3.2 in the 1D case, where the transport map (after removing the common mass) is decreasing as a consequence of Remark 2.12. Note that the precise structure of the optimal transport map for this kind of "concave" costs in 1D follows a sort of hierarchical construction, first investigated in [230] and then used in [142] for numerical purposes.

This suggests to study the case where μ and ν are mutually singular, and the best that one could do would be to prove the existence of an optimal map in this case. In particular, this allows to avoid the singularity of the function $(x,y) \mapsto \ell(|x-y|)$ concentrated on the diagonal $\{x = y\}$ (look at the example $|x-y|^\alpha$), since when the two measures have no common mass almost no point x is transported to $y = x$.

Yet, exploiting this fact needs some attention. The easiest case is when μ and ν have disjoint supports, since in this case there is a lower bound on $|x-y|$ and this allow to stay away from the singularity. Yet, $\mathrm{spt}(\mu) \cap \mathrm{spt}(\nu) = \emptyset$ is too restrictive, since even in the case where μ and ν have smooth densities f and g it may happen that, after subtracting the common mass, the two supports meet on the region $\{f = g\}$.

The problem has been solved in the classical paper by Gangbo and McCann, [177], one of the first classical papers about optimal transportation. In such a paper, the authors used the slightly less restrictive assumption $\mu(\mathrm{spt}(\nu)) = 0$. This assumption covers the example above of two continuous densities, but does not cover many other cases. Think at μ being the Lebesgue measure on a bounded domain Ω and ν being

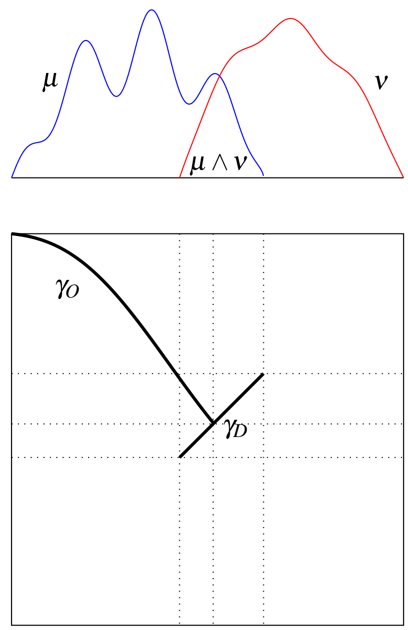

Figure 3.5: The optimal plan in 1D in the case of common mass

an atomic measure with an atom at each rational point, or other examples that one can build with fully supported absolutely continuous measures concentrated on disjoint sets A and $\Omega \setminus A$ (see **Ex**(20)). In the present section we show how to prove the existence of a transport map when μ and ν are singular to each other and $\mu \ll \mathscr{L}^d$.

We will prove the following result.

Theorem 3.26. *Suppose that μ and ν are two mutually singular probability measures on $\mathbb{R}^d$ such that $\mu \ll \mathscr{L}^d$, and take the cost $c(x,y) = \ell(|x-y|)$, for $\ell : \mathbb{R}_+ \to \mathbb{R}_+$ strictly concave, C^1 on $]0,+\infty[$, and increasing. Then there exists a unique optimal transport plan γ and it is induced by a transport map.*

Proof. We will use the standard procedure explained in Chapter 1 for costs of the form $c(x,y) = h(x-y)$. In particular we get the existence of a Kantorovich potential φ such that, if $(x_0,y_0) \in \mathrm{spt}(\gamma)$, then

$$x_0 - y_0 = (\nabla h)^{-1}(\nabla \varphi(x_0)). \tag{3.8}$$

This allows to express y_0 as a function of x_0, thus proving that there is only one point $(x_0,y) \in \mathrm{spt}(\gamma)$ and hence that γ comes from a transport $\mathrm{T}(x) = x - (\nabla h)^{-1}(\nabla \varphi(x))$. This approach also proves uniqueness in the same way. In Chapter 1 we presented it under strict convexity assumptions on h, so that ∇h is injective and $(\nabla h)^{-1} = \nabla h^*$. But the injectivity is also true if $h(z) = \ell(|z|)$. Indeed we have $\nabla h(z) = \ell'(|z|)\frac{z}{|z|}$, and the modulus of this vector identifies the modulus $|z|$ (since ℓ' is strictly increasing) and the direction identifies the direction of z.

The main difficulty is the fact that we need to guarantee that φ is differentiable a.e. with respect to μ. Since $\mu \ll \mathscr{L}^d$, it is enough to have φ Lipschitz continuous, which is usually proven using the fact that $\varphi(x) = \inf_y h(|x-y|) - \psi(y)$. Yet, concave functions on $\mathbb{R}^+$ may have an infinite slope at 0 and be non-Lipschitz, and this could be the case for φ as well. This suggests the use of an alternate notion of differentiability.

Box 3.3. **– Important notion** – Approximate gradient

We recall here some facts about a measure-theoretical notion replacing the gradient for less regular functions. The interested reader can find many details in [161].

Let us start from the following observation: given a function $f : \Omega \to \mathbb{R}$ and a point $x_0 \in \Omega$, we say that f is differentiable at $x_0 \in \Omega$ and that its gradient is $\mathbf{v} = \nabla f(x_0) \in \mathbb{R}^d$ if for every $\varepsilon > 0$ the set

$$A(x_0, \mathbf{v}, \varepsilon) := \{x \in \Omega : |f(x) - f(x_0) - \mathbf{v} \cdot (x - x_0)| > \varepsilon |x - x_0|\}$$

is at positive distance from x_0, i.e. if for small $\delta > 0$ we have $B(x_0, \delta) \cap A(x_0, \mathbf{v}, \varepsilon) = \emptyset$. Instead of this requirement, we could ask for a weaker condition, namely that x_0 is a zero-density point for the same set $A(\mathbf{v}, \varepsilon)$ (i.e. a Lebesgue point of its complement). More precisely, if there exists a vector $\mathbf{v}$ such that

$$\lim_{\delta \to 0} \frac{|A(x_0, \mathbf{v}, \varepsilon) \cap B(x_0, \delta)|}{|B(x_0, \delta)|} = 0$$

then we say that f is approximately differentiable at x_0 and its approximate gradient is $\mathbf{v}$. The approximate gradient will be denoted by $\nabla_{\mathrm{app}} f(x_0)$. As one can expect, it enjoys several of the properties of the usual gradient, that we list here.

- The approximate gradient, provided it exists, is unique.
- The approximate gradient is nothing but the usual gradient if f is differentiable.
- The approximate gradient shares the usual algebraic properties of gradients, in particular $\nabla_{\mathrm{app}}(f+g)(x_0) = \nabla_{\mathrm{app}} f(x_0) + \nabla_{\mathrm{app}} g(x_0)$.
- If x_0 is a local minimum or local maximum for f, and if $\nabla_{\mathrm{app}} f(x_0)$ exists, then $\nabla_{\mathrm{app}} f(x_0) = 0$.

Another very important property is a consequence of Rademacher theorem.

Proposition - Let $f, g : \Omega \to \mathbb{R}$ be two functions defined on a same domain Ω with g Lipschitz continuous. Let $E \subset \Omega$ be a Borel set such that $f = g$ on E. Then f is approximately differentiable almost everywhere on E and $\nabla_{\mathrm{app}} f(x) = \nabla g(x)$ for a.e. $x \in E$.

Proof - It is enough to consider all the Lebesgue points of E where g is differentiable. These points cover almost all E. It is easy to check that the definition of approximate gradient of f at a point x_0 is satisfied if we take $\mathbf{v} = \nabla g(x_0)$.

As a consequence, it is also clear that every countably Lipschitz function is approximately differentiable a.e.

We just need to prove that φ admits an approximate gradient Lebesgue-a.e.: this would imply that Equation (3.8) is satisfied if we replace the gradient with the approximate gradient.

Recall that we may suppose

$$\varphi(x)=\varphi^{cc}(x)=\inf_{y\in\mathbb{R}^d}\ell(|x-y|)-\varphi^c(y)\ .$$

Now consider a countable family of closed balls B_i generating the topology of $\mathbb{R}^d$, and for every i consider the function defined as

$$\varphi_i(x):=\inf_{y\in B_i}\ell(|x-y|)-\varphi^c(y)\ .$$

for $x\in\mathbb{R}^d$. One cannot provide straight Lipschitz properties for φ_i, since a priori y is arbitrarily close to x and in general ℓ is not Lipschitz close to 0. However φ_i is Lipschitz on every B_j such that $\operatorname{dist}(B_i,B_j)>0$. Indeed if $x\in B_j$, $y\in B_i$ one has $|x-y|\geq d>0$, therefore the Lipschitz constant of $\ell(|\cdot-y|)-\varphi^c(y)$ does not exceed $\ell'(d)$. It follows that φ_i is Lipschitz on B_j, and its Lipschitz constant does not exceed $\ell'(d)$.

Then φ has an approximate gradient almost everywhere on $\{\varphi=\varphi_i\}\cap B_j$. By countable union, φ admits an approximate gradient a.e. on

$$\bigcup_{\substack{i,j\\ d(B_i,B_j)>0}}[\{\varphi_i=\varphi\}\cap B_j]\ .$$

In order to prove that φ has an approximate gradient μ-almost everywhere, it is enough to prove that

$$\mu\left(\bigcup_{\substack{i,j\\ d(B_i,B_j)>0}}\{\varphi_i=\varphi\}\cap B_j\right)=1.$$

In order to do this, note that for every i and j we have

$$\pi_x(\operatorname{spt}(\gamma)\cap(B_j\times B_i))\subset\{\varphi=\varphi_i\}\cap B_j\ .$$

Indeed, let $(x,y)\in\operatorname{spt}(\gamma)\cap(B_j\times B_i)$. Then $\varphi(x)+\varphi^c(y)=l(|x-y|)$. It follows that

$$\varphi_i(x)=\inf_{y'\in B_i}\ell(\left|x-y'\right|)-\varphi^c(y')\leq\ell(|x-y|)-\varphi^c(y)=\varphi(x)\ .$$

On the other hand, for every $x\in\mathbb{R}^d$

$$\varphi_i(x)=\inf_{y\in B_i}\ell(|x-y|)-\varphi^c(y)\geq\inf_{y\in\mathbb{R}^d}\ell(|x-y|)-\varphi^c(y)=\varphi(x)\ .$$

As a consequence of this,

$$\begin{aligned}\mu\left(\bigcup_{\substack{i,j\\ d(B_i,B_j)>0}}\{\varphi_i=\varphi\}\cap B_j\right)&\geq\mu\left(\bigcup_{\substack{i,j\\ d(B_i,B_j)>0}}\pi_x(\operatorname{spt}(\gamma)\cap(B_j\times B_i))\right)\\ &=\mu\left(\pi_x\left(\operatorname{spt}(\gamma)\cap\bigcup_{\substack{i,j\\ d(B_i,B_j)>0}}B_j\times B_i\right)\right)\\ &=\mu(\pi_x(\operatorname{spt}(\gamma)\setminus D))\\ &=\gamma\left[(\pi_x)^{-1}(\pi_x(\operatorname{spt}(\gamma)\setminus D))\right]\\ &\geq\gamma(\operatorname{spt}(\gamma)\setminus D)=1,\end{aligned}$$

where D is the diagonal in $\Omega \times \Omega$. □ □

From the previous theorem, we can also easily deduce the following extension. Define $\mu \wedge \nu$ as the maximal positive measure which is both less or equal than μ and than ν, and $(\mu - \nu)_+ = \mu - \mu \wedge \nu$, so that the two measures μ and ν uniquely decompose into a common part $\mu \wedge \nu$ and two mutually singular parts $(\mu - \nu)_+$ and $(\nu - \mu)_+$.

Theorem 3.27. *Suppose that μ and ν are probability measures on Ω with $(\mu - \nu)_+ \ll \mathscr{L}^d$, and take the cost $c(x,y) = \ell(|x-y|)$, for $\ell : \mathbb{R}_+ \to \mathbb{R}_+$ strictly concave, C^1 on $]0,+\infty[$, and increasing with $\ell(0) = 0$. Then there exists a unique optimal transport plan γ, and it has the form* $(\mathrm{id},\mathrm{id})_\#(\mu \wedge \nu) + (\mathrm{id},\mathrm{T})_\#(\mu - \nu)_+$.

As we said, the above results can be extended to more general situations: differentiability of ℓ is not really necessary, and the assumption $\mu \ll \mathscr{L}^d$ can be weakened. In [250], the result is proven under the natural assumption that μ does not give mass to small sets, i.e. for every $A \subset \mathbb{R}^d$ which is $(d-1)$-rectifiable we have $\mu(A) = 0$. The key tool is the following lemma, which is an interesting result from Geometric Measure Theory that can be used instead of Lebesgue points-type results when we face a measure which is not absolutely continuous but "does not give mass to small sets". It states that, in such a case, for μ-a.e. point x every cone exiting from x, even if very small, has positive μ mass. In particular it means that we can find points of $\mathrm{spt}(\mu)$ in almost arbitrary directions close to x. This lemma is strangely more known among people in optimal transport (see also [122]), than in Geometric Measure Theory (even if it corresponds more or less to a part of Lemma 3.3.5 in [162]).

Lemma 3.28. *Let μ be a Borel measure on $\mathbb{R}^d$, and suppose that μ does not give mass to small sets. Then μ is concentrated on the set*

$$\{x : \forall \varepsilon > 0, \forall \delta > 0, \forall e \in \mathbb{S}^{d-1},\ \mu(C(x,e,\delta,\varepsilon)) > 0\}\ ,$$

where

$$C(x,e,\delta,\varepsilon) = C(x,e,\delta) \cap \overline{B(x,\varepsilon)} := \{y : \langle y-x, e\rangle \geq (1-\delta)\,|y-x|\} \cap \overline{B(x,\varepsilon)}$$

Chapter 4

Minimal flows, divergence constraints and transport density

4.1 Eulerian and Lagrangian points of view

We review here the main languages and the main mathematical tools to describe static or dynamical transport phenomena.

4.1.1 Static and Dynamical models

This section presents a very informal introduction to the physical interpretation of dynamical models in optimal transport.

In fluid mechanics, and in many other topics with similar modeling issues, it is classical to consider two complementary ways of describing motions, which are called Lagrangian and Eulerian.

When we describe a motion via Lagrangian formalism we give "names" to particles (using either a specific label, or their initial position, for instance) and then describe, for every time t and every label, what happens to that particle. "What happens" means providing its position and/or its speed. Hence we could for instance look at trajectories $y_x(t)$, standing for the position at time t of particle originally located at x. As another possibility, instead of giving names we could consider bundles of particles with the same behavior and indicate how many are they. This amounts to giving a measure on possible behaviors.

The description may be more or less refined. For instance if one only considers two different times $t = 0$ and $t = 1$, the behavior of a particle is only given by its initial and final positions. A measure on those pairs (x, y) is exactly a transport plan. This explains why we can consider that the Kantorovich problem is expressed in Lagrangian coordinates. The Monge problem is also Lagrangian, and particles are labelled by their initial position.